

Expression	Reading	Notes
$A_{m \times n}$	an m -by- n matrix A	Note the position of the dimensions before the matrix itself.
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	a two-by-two matrix, (where) the first row is 1, (and) 2, the second row is 3, (and) 4 a two-by-two matrix, (where) the first column is 1, (and) 3, the second column is 2, (and) 4	For larger matrices simply add more columns/rows.
$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$	an m -by- n matrix, the first row is a sub one one, a sub one two, dot dot dot , a sub one n , second row a sub two one, a sub two two, dot dot dot , a sub two n , third row dot dot dot , m th (or last) row a sub m one, a sub m two, dot dot dot , a sub m n	
A^T	A transpose the transpose of A	It is not usual to say $*A$ <i>transposed</i> which would imply the action that happened to the matrix.
$\det(A)$ or $ A $	the determinant of A	It is always the determinant, as the number is uniquely associated with each matrix.
A^{-1}	A inverse the inverse of A the inverse matrix A	
a_{ij}	a sub i j the ij -th entry/element of matrix A	
Expression	Reading	Notes
$a + b = c$	a plus b is equal to/ equals c the sum of a and b is/ is equal to/equals c	Note <i>Let $a + b = c$ be ...</i> is read as “Let a plus b equal to c be ...”, i.e. dropping “is” when used as a way of rewriting a given quantity as a sum or vice versa (the equality is a fact accomplished rather than an active assertion). A similar remark applies to $a - b = c$, $ab = c$, $a/b = c$.
$2 + 3 = 5$	two plus three equals/makes five two added to three makes five the sum of two and three equals five	And the reverse order, e.g. five is the sum of two and three. <i>Two plus three is five is less formal but acceptable;</i> <i>two and three is/are five is less formal still and not sufficiently clear without context.</i>
$a - b = c$	a minus b is equal to/ equals/makes c a take b is equal to/ equals c	The first is most common; the second less formal.

	the difference of a and b is/ is equal to/ equals c	“the difference between” two numbers is also possible.
$5 - 3 = 2$	five minus three equals/makes two three from five leaves two five diminished by three is equal to two two is/equals the difference of five and three	Less formal but acceptable is <i>five take three is two</i> (or <i>five take three equals/is equal to two</i>). The “diminished by” formulation is uncommon.
$ab = c$	a times b equals c a multiplied by b equals c <i>the product of a and b is/ is equal to/ equals c</i> ab equals c	
$2 \cdot 3 = 6$	two times three equals/makes six two multiplied by three equals six the product of two and three is/ is equal to/ equals six	
$a \div b = c$	a divided by b equals c	
$6 \div 3 = 2$	six divided by three equals/makes two	
$\frac{a}{b}$	(the fraction) a over b ; a divided by b	<i>fraction of ...</i> is incorrect
$\frac{a+b}{c+d}$	(the fraction equal to) (the sum/ the quantity) a plus b over/ divided by (the sum/ the quantity) c plus d ; the fraction whose numerator is ... and whose denominator is ...	If the numerator A in A/B is a lengthy expression, read as “ A , all (that) over B ” or “ A , that quantity (then) divided by B ”. The correct terms <i>numerator</i> and <i>denominator</i> are sometimes erroneously called the <i>*nominator</i> (=a person who nominates) and <i>*denumerator</i> (=nonsense).
$0.\bar{3}$	zero point three recurring/repeating	We do not say <i>*periodic</i> .
a^n	a to the n ; a to the n -th power; a (raised) to the power (of) n ; the n -th power of a	a to the n -th is incorrect, and might in any case be confused with <i>the n-th root of a</i> (“one n th” being $1/n$); a to n is incorrect.
a^0	a to the zero; a to the zeroth power; a (raised) to the power (of) zero; the zeroth power of a	
a^2	a squared, the square of a ; a to the two, a (raised) to the power (of) two; a (raised) to the second power; the second power of a	

a^3	a cubed, the cube of a ; a to the three, a (raised) to the power (of) three; a (raised) to the third power; the third power of a	
$\sqrt[n]{a}$	the n -th root of a	Equals $a^{\frac{1}{n}}$ read as “ a to the one over n ”, “ a to the one n th” etc.
$\sqrt{a}$	root a ; the square root of a	<i>the second root of ...</i> - this phrasing is not used with this meaning.
$\sqrt[3]{a}$	the cube root of a ; the third root of a	<i>the third root of</i> is uncommon but understood in the required sense of denoting the cube root. For higher roots, <i>the fourth root of a</i> , <i>the fifth root of a</i> , etc. <u>are</u> used, in line with “the n -th root of a ”.
Expression	Reading	Notes
$a \in A$	a is an element of A a belongs to A	Written after an equation this could be read <i>where a belongs to A</i>, or <i>for a from A</i>. Example: $0^n = 0, n \in \mathbb{N}$ <i>...where n is a natural number</i> <i>...where n is natural</i> <i>...for natural n</i> <i>...for n from the set of natural numbers</i>
$A \subset B$	A is a subset of B A is contained in B	The article a is very important here.
$\forall$	for all for every	In various contexts it is also possible to use <i>for any</i>, <i>for each</i>.
$\forall a \in \mathbb{R}$	for all real a for all a from the set of real numbers	
$\exists$	there exist(s) there is/are	Not enough to say just <i>*exist(s)</i>, the grammar requires some subject.
$p \wedge q$	p and q the conjunction of p and q	The conjunction/disjunction option is quite infrequent.
$p \vee q$	p or q the disjunction of p and q	
$p \Rightarrow q$	if p , then q p implies q	Make sure you use the <i>if...then...</i> structure, which is the most common way of reading implications (for example in theorems), e.g. <i>if a is greater than 2, then a is greater than 3</i>.
$p \Leftrightarrow q$	p if and only if q p is equivalent to q	In writing, this might be informally shortened to <i>iff</i>.
$\neg p$	not p the negation of p	

$A \cap B$	the intersection of A and B A intersect B	
$A \cup B$	the union of A and B A union B	
$C_U(A)$	the complement of A with respect to U the complement of A in U	It is usual that the set with respect to which we take the complement is only understood and not explicitly stated, see below.
$C(A)$ or A'	the complement of A	
$A \setminus B$	A minus B the difference A minus B the difference of A and B	
:	such that („takové, že“) or it holds that („platí, že“)	The meaning of this symbol depends on the context. Two examples below.
$\forall a \exists b : a \cdot b = 0$	for all a there exists b such that a times b equals 0	
$\forall A : A \cap \emptyset = \emptyset$	for all A it holds that A intersect empty set is an empty set	In this context it is perfectly fine to not read the symbol at all, i.e. <i>for all A, A intersect the empty set is the empty set.</i>
Expression	Reading	Notes
$\{a_n\}$	the sequence a sub n	The <i>n-th element</i> of the sequence would be <i>a sub n</i>.
(a, b)	the open interval a b	The notation can refer to other objects than intervals, e.g. <i>ordered pair</i>.
$[a, b]$	the closed interval a b	For all intervals, <i>a</i> and <i>b</i> are called the <i>endpoints</i>. It is <i>the interval with endpoints a, b</i>.
$[a, b)$ $(a, b]$	the half-closed interval, open on the left/right and closed on the right left	
$f(x)$	f of x function f of x	This structure works for all functions: the [function] of [argument] just insert any specific function, e.g. trigonometric functions (the sine of x), logarithms, roots, derivative.
$f^{-1}(x)$	the inverse of f of x the inverse of the function f of x	
$\int f(x)dx$	the (indefinite) integral of f of x with respect to x	The word <i>definite/indefinite</i> can be left out, it is clear from the context.
$\int_a^b f(x)dx$	the (definite) integral from a to b of f of x with respect to x the (definite) integral of f of x from a to b with respect to x	<i>with respect to x</i> can be replaced with <i>d x</i> at any time.

$+ c$	plus c plus constant	this is the <i>constant of integration</i>.
$\lim_{x \rightarrow \infty} f(x)$	the limit of f of x as x approaches infinity the limit of f of x as x goes to infinity the limit of f of x as x tends to infinity	- The order of the function (<i>of f of x</i>) and where it approaches (<i>as x goes to ...</i>) can be switched. - The three verbs are interchangeable.
$\lim_{x \rightarrow 0^-} f(x)$	the limit of f of x as x approaches zero from the left the limit of f of x as x approaches zero from below the left-handed limit of f of x as x approaches zero	
$\lim_{x \rightarrow 0^+} f(x)$	the limit of f of x as x approaches zero from the right the limit of f of x as x approaches zero from above the right-handed limit of f of x as x approaches zero	
$\sum_{n=1}^{\infty} a_n$	the sum of a sub n from n equals 1 to infinity	
$f'(x)$	the (first) derivative of f of x f prime of x	The <i>first</i> can be omitted, it is redundant unless you want to stress this fact.
$f'(a)$	the derivative of f at a	This is the derivative at a point – mind the preposition.
or $\frac{df}{dx}$ $\frac{d}{dx}f(x)$	the (first) derivative of f of x (with respect to x) df by dx d by dx of f of x	- This means the same as the above; so called ‘Leibnizian notation’, still used in the Anglo-Saxon context. Expresses explicitly what you differentiate with respect to. - extra example: $\frac{d}{dx} \sin x$ “d by dx of sine x”
$f''(x)$	the second derivative of f of x f double prime of x	For derivatives of higher order: <i>the third/fourth... derivative ...</i> <i>f triple prime ...</i>

$\sin x$	the sine of x sine x	- Do not say <i>*sin x</i>. - Note that the names of the trigonometric functions in English do not derive from Latin (as they do in Czech), so no <i>*sinus</i>, <i>*cosinus</i>, <i>*tangens</i>.
$\cos x$	the cosine of x cosine x cos x	
$\tan x$	the tangent of x tangent x tan x	
$\cot x$	the cotangent of x cotangent x cot x	
$\sin^{-1} x$ $\cos^{-1} x$ $\tan^{-1} x$ $\cot^{-1} x$	the arc sine/cosine/... of x the inverse sine/cosine/... of x	- It is not a power, so do not say <i>*sine to the minus one</i>. - It is preferred to avoid <i>*arcus</i> (as it comes from Latin). - The spelling <i>arcsine</i> and <i>arc-sine</i> is also correct.
$\log_a b$	the logarithm to (the) base a of b the logarithm of b to/with (the) base a log-a of b the base-a logarithm of b	
$\log x$	the (common /decimal/decadic) logarithm of x the logarithm to (the) base 10 of x the logarithm of x to/with (the) base 10 log-ten of x the base-10 logarithm of x	It is useful to remember that it is called the <i>common</i> logarithm.
$\ln x$	the natural logarithm of x the logarithm to the base e of x	